QP CODE: 25900222

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) -MG2DSCZIM101-
PUBLIC HEALTH MICROBIOLOGY
(2024 Admission onwards)

Duration: 1.5 Hours **Max.Marks: 50**

Part A
Fill in the blanks,
Answer **all** questions. Each question carries **1** mark

1. …... is the boiling point of water [K]/[CO4]
   Ans: 100℃
2. HEPA Filter remove particles as small as …… diameter [K]/[CO2]
   Ans: 0.3
3. …... is the incubation period of bacillus dysentery. [U]/[CO5]
   Ans: 1-3 days
4. Candling is the method to detect …… [U]/[CO5]
   Ans: Egg spoilage
5. …... defines Public health is health state of complete physical, mental and social well being. [U]/[CO1]
   Ans: Public health
6. ….. is known as bakers yeast [U]/[CO5]
   Ans: *Saccharomyces cervisiae*
7. …. is measuring unit of noise pollution. [K]/[CO1]
   Ans: Decibels
8. …. is the incineration temperature. [K]/[CO5]
   Ans: 1200℃
9. MF Broth stands for …. [K]/[CO4]
   Ans: Membrane filter endo broth
10. Water borne diseases are …. [K]/[CO5]
    Ans: Cholera, Bacillus dysentery

(1 x 10 = 10 Marks)

Part B

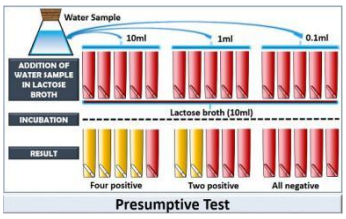
Short answer questions

Answer any **10** questions. Each question carries **2** marks

11. List the materials required for sampling of bacteria in the air. [K]/[CO2]

Air sampler, Media, petridish, incubator,colony counter etc

12. Draw the presumptive test. [U]/[CO4]



13. Describe staphylococcal food poisoning. [K]/[CO5]

It is a type of foodborne illness caused by eating food contaminated with toxins produced by *Staphylococcus aureus* bacteria.

14. Summarize prevention methods in milk. [K]/[CO5]

Pasteurization, boiling, Drying, Freezing, Refrigeration, sterilization etc.

15. Discuss effects of air pollution. [K]/[CO1]

Human health effects, environmental effects, climate change, economic impact

16. Identify symptoms of pneumonia. [U]/[CO3]

Cough,Fever, Shortness of Breath,Chest Pain,Fatigue,Rapid or Difficulty Breathing,Bluish Lips or Fingertips,Loss of Appetite,Nausea, Vomiting, or Diarrhea, ` Confusion or Delirium

17. Describe the laboratory diagnosis of typhoid. [K]/[CO5]

Rapid Diagnostic Tests (RDTs) ,Blood Culture (Gold Standard),Widal test.

18. Define food hygiene. [U]/[CO5]

It refers to the condition and practices necessary to ensure the safety of food from production to consumption,prevention contamination & foodborne illness.

19. Justify health effects of air pollution. [K]/[CO2]

Respiratory Diseases,Cardiovascular Diseases, Neurological Diseases.

20. List out the enzymes involved in fish spoilage. [U]/[CO5]

TMAO Reductase,Proteases ,Lipases,Urease,Deaminases,Decarboxylases.

21. Classify bacterial waterborne diseases. [U]/[CO4]

Cholera, Typhoid Fever, Dysentery (Amoebic & Bacillary)

22. State the transmission methods of water contamination. [U]/[CO4]

Sewage Discharge, Industrial Waste, Agricultural Runoff, Stormwater Runoff, Leaking Septic System, Landfills, Mining Activities

(10 x 2 = 20 Marks)

Part C

Short Essay Questions

Answer any **5** questions. Each question carries **4** marks

23. Illustrate the cholera pandemics. [K]/[CO4]

(Introduction-2 marks, History- 2 marks)

It is an acute diarrhoeal diseases. This pandemic was caused not by the classical *cholera vibrio*, but by *El Tor vibrio*. This produced milder disease and mortality was minimal.

24. Categorize symptoms of HAV. [K]/[CO4]

(Types-2 marks, Symptoms- 2 marks)

Fever, Fatigue ,Loss of appetite, Nausea and vomiting, Abdominal pain ,Dark-colored urine, Pale or clay-colored stools, Joint pain,Jaundice (yellowing of skin and eyes),Itchy skin

25. Write about waste disposal. [U]/[CO4]

(Techniques of waste disposal & Types-1 marks each)

Incineration, deep burial, chemical treatment, autoclaving, sewage drains

26. List the types of samples should be collected in water borne diseases. [U]/[CO4]

(Types of samples-1 marks each)

Water samples, Stool samples, Blood samples, Urine samples & tissue samples.

27. Explain the properties of a good disinfectant. [K]/[CO4]

(Definition- 1 mark, Properties- 3 mark)

Active against a variety of microorganisms, soluble in water, without unpleasant odour, stable on storage at room temperature, inexpensive.

28. State the symptoms of Tuberculosis and measles. [K]/[CO3]

(Symptoms of Tuberculosis-2 mark, Symptoms of measles-2 mark)

Tuberculosis: cough, chest pain,fatigue, phlegm, swollen lymph nodes.

Measles: fever, cough, conjunctivitis and rash

29. Evaluate major environmental impact on human health. [K]/[CO2]

(Types & impact- 1 mark each)

Air pollution, water pollution, Radiation, Climate change, Noise pollution,soil pollution.

(5 x 4 = 20 Marks)